

KHUJAND, Tajikistan

WMO: 385990

Lat: 40.215N

Lon: 69.695E

Elev: 1401

StdP: 13.97

Time Zone: 5.00 (E05)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	19.8	9.7	9.5	22.7	14.1	11.8	28.1	27.9	35.6	26.4	34.7	11.8	250	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	100.8	66.4	98.7	66.1	96.6	65.3	69.5	92.7	68.2	91.1	67.2	89.9	11.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	89.7	74.6	60.9	84.1	74.2	59.2	79.0	74.6	34.4	93.7	33.3	91.2	32.4	89.4	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	23.5	21.7	13.3	106.0	8.3	1.9	7.4	107.4	2.5	108.5	-2.1	109.6	-8.1	111.0
			12.5	72.4	8.2	1.3	6.7	73.3	1.9	74.1	-2.7	74.9	-8.6	75.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.1	34.1	37.8	51.5	62.2	72.9	81.4	84.2	80.9	72.0	59.0	46.3	36.8	(6)
(7)		DBStd	18.86	6.96	8.84	7.85	7.54	5.54	4.76	3.97	4.59	5.48	7.55	7.63	6.31	(7)
(8)		HDD50	1502	492	345	78	8	0	0	0	0	0	16	153	410	(8)
(9)		HDD65	3935	957	761	419	136	6	0	0	0	10	212	560	874	(9)
(10)		CDD50	5174	0	4	125	374	711	942	1059	958	661	296	43	1	(10)
(11)		CDD65	2129	0	0	1	51	252	492	594	493	220	26	0	0	(11)
(12)		CDH74	25814	0	0	19	513	2482	6175	7997	6134	2258	233	3	0	(12)
(13)		CDH80	13526	0	0	2	144	1042	3385	4648	3340	920	45	0	0	(13)
(14)	Wind	WSAvg	10.6	11.8	12.4	11.6	10.7	9.8	9.9	10.5	9.8	9.4	9.5	9.8	11.6	(14)
(15)	Precipitation	PrecAvg	6.4	0.6	0.7	1.0	1.1	0.8	0.2	0.1	0.0	0.1	0.6	0.7	0.7	(15)
(16)		PrecMax	13.0	2.3	1.9	2.9	3.3	2.7	0.9	0.9	0.4	0.9	3.5	2.4	2.4	(16)
(17)		PrecMin	3.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.2	0.5	0.4	0.7	0.8	0.6	0.2	0.2	0.1	0.2	0.7	0.5	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	63.4	75.8	89.2	96.7	104.0	105.4	102.2	95.4	84.5	72.2	55.4	(19)
(20)			MCWB	42.2	48.4	55.4	61.1	62.9	65.6	67.5	67.6	63.4	59.6	55.9	45.8	(20)
(21)		2%	DB	48.0	55.4	71.3	84.3	93.1	100.2	102.1	98.9	91.9	80.3	66.4	50.3	(21)
(22)			MCWB	41.1	45.7	54.1	59.8	62.5	65.1	67.0	66.7	62.7	59.0	53.2	42.5	(22)
(23)		5%	DB	44.8	51.8	67.7	80.2	89.6	97.0	99.0	96.8	89.1	75.6	62.1	47.8	(23)
(24)			MCWB	39.6	43.8	52.9	58.9	61.6	64.5	66.5	66.0	62.1	56.4	50.6	41.2	(24)
(25)		10%	DB	42.7	48.6	64.1	75.5	86.1	94.7	96.9	94.6	85.7	71.9	57.5	44.9	(25)
(26)			MCWB	38.4	42.3	51.7	57.3	61.5	63.9	65.8	65.7	60.8	55.2	48.1	40.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	50.2	58.1	63.9	66.3	69.2	71.6	71.1	66.6	61.7	57.2	46.8	(27)
(28)			MCD B	48.1	58.0	71.1	77.9	85.6	88.8	95.9	92.1	86.6	78.0	70.8	52.4	(28)
(29)		2%	WB	42.0	47.3	56.1	62.3	64.9	67.6	69.9	69.4	65.1	60.0	54.4	44.1	(29)
(30)			MCD B	45.9	53.9	66.5	76.0	82.7	89.1	93.9	92.4	85.7	75.3	64.0	47.9	(30)
(31)		5%	WB	40.2	45.0	54.5	61.0	63.8	66.6	68.5	68.0	63.5	58.6	51.9	42.2	(31)
(32)			MCD B	43.9	50.1	64.1	74.4	80.8	88.3	92.1	91.1	83.9	72.8	59.5	45.8	(32)
(33)		10%	WB	38.3	43.1	53.0	59.4	63.0	65.7	67.4	66.8	62.1	57.1	49.6	40.4	(33)
(34)			MCD B	41.7	47.4	62.0	71.9	79.3	87.3	90.7	89.6	81.9	68.6	56.2	43.9	(34)
(35)	Mean Daily Temperature Range	MDBR	10.1	12.3	17.3	20.7	22.5	24.3	24.5	25.1	25.4	20.9	15.7	10.8	(35)	
(36)		5% DB	MCD BR	12.7	15.9	23.0	26.9	28.0	28.4	28.0	28.5	29.8	25.6	21.3	14.3	(36)
(37)			MCWBR	7.6	8.6	9.9	9.1	7.5	8.0	8.7	9.6	9.9	9.8	10.0	8.3	(37)
(38)		5% WB	MCD BR	11.7	13.8	19.8	21.6	23.2	24.4	25.6	26.2	27.3	22.5	17.8	12.5	(38)
(39)			MCWBR	7.7	8.4	9.3	8.2	7.2	7.4	8.7	9.4	9.8	8.7	9.2	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.364	0.414	0.479	0.498	0.494	0.493	0.479	0.453	0.439	0.477	0.436	0.368	(40)	
(41)		taud	2.153	2.031	1.916	1.920	1.984	1.992	2.032	2.094	2.068	1.913	1.981	2.125	(41)	
(42)		Ebn at Noon	242	245	242	248	252	252	254	258	252	220	210	228	(42)	
(43)		Edh at Noon	34	44	55	59	56	56	53	49	47	49	40	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	624	875	1239	1632	2029	2347	2377	2166	1769	1153	707	537	(44)	
(45)		RadStd	61	110	114	180	130	100	83	100	59	113	91	68	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	+151	+115	(47)

Nomenclature: See separate page